DISSERTATION PROPOSAL

Camera Trap Data Analysis

in collaboration with ENV

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1. Background and Description of Project

As we all know, living beings and their surroundings massively destruct wildlife and their environment, and hence to prevent further extinction of wildlife, it is very much important to track and record their information closely. Unfortunately, humans do not have both the time and availability to fulfil this important requirement of ecology. As of now, most of the capturing of animals, their movements, and different species kingdom is done through cameras only.

Cameras are considered as one of the most functional weapons whenever the topic is about capturing movement of wildlife. One can extract useful information of particular species such as activity patterns, species habitat, and ecological interaction as well. As of now, traditional camera trap systems based on 2D imaging provide limited functionality, which is considered a drawback while capturing detailed observation. Such drawbacks mostly affect wildlife well-being evaluation and individual recognition.

Latest inventions of technology in AI, machine learning, and computer vision can be used as a solution to overcome these difficulties. The use of 3D capturing provides an advantage to extract detailed information and layout configuration of a particular spot. These features present objects in three dimensions, through which one can gain essential knowledge like distance and depth from the camera, along with accurate shape of the body with improved AI learning.

This research and its expected accurate outcomes are highly important for wildlife habitats and for their preservation. After discovering the capabilities of 3D imaging, it can be appreciated that this technology has the power to overcome the difficulties of capturing wildlife and preserve their future in a better way. The verdict of this study contributes as a precious groundwork for future ecological studies and wildlife management strategies.

2. Aims and Objectives

This dissertation includes a major task, which is to build a camera trap data analysis system after combining 3D imaging with AI-driven computer vision technology, which only focuses on improvisation in capturing wildlife. Another important aim of this research is to investigate and overcome the restrictions caused by traditional cameras while capturing wildlife through developing various technology-enhanced models. These models not only help in improvisation in capturing wildlife but also contribute to discovering new ways for prevention of wildlife.

2.1 Aims

- To investigate the effectiveness of integrating 3D imaging techniques with Artificial Intelligence and computer vision for enhancing wildlife monitoring and analysis.
- To develop an intelligent and automated system for analysing camera trap data to improve species identification, behavioural analysis, and ecological insights.
- To analyse wildlife population dynamics, including species distribution, abundance, and rarity, using camera trap data.
- To establish a standardised approach for camera trap metadata collection to improve interoperability with global biodiversity platforms such as Global Biodiversity Information Facility (GBIF) and Wildlife Insights.
- To extract detailed wildlife information, including movement patterns, activity trends, spatial behaviour, and (where applicable) depth and size estimation using advanced imaging techniques.

2.2 Objectives

- Development of 3D camera imaging and make comparison on the basis of accuracy in analysis, data representation, and distance management with a 2D camera system. This will give a clear classical brief difference on why the 3D camera trap processes data better than traditional cameras.
- Machine learning models such as CNN, which perform self-regulating spotting and identification of wildlife species, are going to be developed and implemented within this study.
- Evaluation of current 3D technologies through which integration can be performed with camera trap systems and also assess their appropriateness for future use.
- To develop a user interface using technologies like Flask, so that any person without technical knowledge can submit information such as images and get access to information such as data reports and prediction review.
- To develop and implement a class of neural network, which is a Siamese network, which compares two inputs of animals for their individual identification.
- Converting images into useful data from raw ones clicked by camera traps after performing preprocessing.

3. Critical Review of Literature

3.1 Traditional Camera Traps and Their Evolution Over Time

Camera traps are the most used gadget since one decade when the topic is about capturing wildlife. This is a global expanding market which currently stands at $392 million (2024), and according to predictions, it will be $815 million (2033). If we take a rough estimation of reports, it is an increase of 108% over the past 9 years. Earlier, there were only very few researchers who used camera traps for prevention of wildlife conservation. There is a mass collection of data which is next to impossible to observe alone and analyse through observation. Although there are numerous limitations of this 2D imaging technology, such as lack of depth information and lack of accuracy in data observation such as size, distance, and movements.

3.2 Limitation of Traditional Camera System

The major cause amongst all the demerits is acquiring limited detailed information such as size and distance from animals. 2D cameras capture two coordinate data (x, y), which are colour and location, but no z-variable for the distance factor. This leads to multiple problems such as inaccurate body measurements, visibility obscurity, and acute distortion. Significant health details like how an animal stands and walks could be essential details for researchers, which are not captured properly by 2D cameras. Along with this, another factor which creates confusion for research is re-identification of the same animal from different angles of the camera when the presence of the animal varies with light and pose. Another drawback of 2D cameras is that their functioning depends on motion sensors to capture images and sense movement; hence all other immaterial motion such as rain, falling leaves, and movements due to wind can be falsely captured by these cameras. Sometimes it may miscapture animal pictures as it starts capturing only when motion gets detected.

3.3 The Evolution of Animal Re-Identification: From Feature Engineering to Metric Learning

Wildlife identification is considered as one of the leading ways for the development of computer vision. In the early era, scientists and researchers used to perform manual comparison of data with a ruleset called feature engineering. Problems were present with this approach such as the need for expert knowledge, the possibility of human error, and specific requirements such as good quality images, good lighting, and animal markings fully visible. For many years, feature engineering was the only technique, which mainly consists of algorithms that do not learn by themselves but only focus on manual instructions, which capture stripes and spots, limiting information for conservationists, researchers, and ecologists.

An advanced improvised AI technique in which neural networks learn by themselves how two images are similar instead of just looking for manually instructed information like stripes and spots is referred to as deep metric learning along with computer vision. This system works on vectors and embedding space. If similarity between animals matches up to a certain extent, then vectors will be placed together; otherwise, they will be placed far from each other. There are two training methods, contrastive and triplet loss, which train models to compare images. Although CNN (Convolutional Neural Network) plays a major role in reading images and converting them into embeddings, which represent a unique digital fingerprint of the animal.

3.4 Depth-Based Welfare Detection: Achievements and Transferability Gap

Depth-based imaging is working diligently in the field of animal welfare within farm animal research, where there is an establishment of controlled barn environments which helps in verification and updating of automatic monitoring systems. Top-view depth cameras are applied to get three-dimensional images, which are further used by systems to get a better understanding of physiological and behavioural metrics. 3D depth images are widely used to detect lameness severity of animals detected by the back picture of the animal. Structural features like the spine and hook bones allow this system to predict any irregularity of movement.

According to a recent study by Tun et al., published in the Journal of Imaging, it has been shown that merging top-view depth cameras with Detectron2, along with Intersection over Union (IoU)-based tracking techniques, provides precision in tracking a particular animal which is already present in a large group by using the back portion of an individual cow. Also, it has been observed that this experiment achieved detection accuracy of 99.94% and a tracking accuracy of 99.92% in just a three-day interval. This is a definition of effectiveness which is gained after combining computer vision technology with AI and depth imaging. These systems are mostly used in live animal farms.

3.5 Toward 3D Feature Integration, Pose Normalisation, and Research Positioning

3D imaging and the use of its features simply rely on the core concept: animals are three-dimensional; hence the capturing of animals must be done by 3D technology only. The issue is that 2D cameras lack performance when the angle and lighting change. Depth-based systems work well but only in controlled environments. Hence, the major point is whether methods which provide performance in controlled environments can work in uncontrolled wildlife.

Nowadays, one of the most significant systems for capturing 3D images is SOCRATES, which is basically a battery-powered stereo camera and an upgraded version for animal detection compared to 2D systems. While SOCRATES does not really capture peak depth information such as shape, keypoints, or identity from the depth data, it clearly shows that 3D capturing cameras exist, but the method to extract 3D features is missing. This is the main gap which this study aims to address.

On the other side, improvisation of pose-based models done by a technique called CKPE leads to better representation of animal keypoints along with Vision Transformer. This helps in the reduction of large datasets. Apart from this, there is another approach which rebuilds animal image shape without the use of hand-tagging after using a 3D Gaussian technique.

To conclude, it has been observed that there is a clear progression from 2D images to acquiring 3D information due to the demerits of using 2D cameras. SOCRATES provides the hardware for capturing 3D data, livestock welfare studies show how to use depth information effectively, and methods like CKPE and Pose Splatter offer advanced ways to extract features with less manual labelling. Nevertheless, all these technologies cannot yet be embedded into one system which can perform for real-world applications. The aim of this dissertation is to develop a system which can be useful for both animal welfare and computer vision technology.

4. Methodology and Methods

4.1 Research Design

This dissertation follows a measurable trial which completely relies on an organized computational pipeline which consists of data preprocessing, 3D feature extraction, model development, and evaluation. This study does not deal with live animals for sourcing of data, as all the data will be obtained from datasets which are publicly available. Execution accuracy of the model depends on standard computer vision metrics; they are recall, precision, mean average precision, and accuracy.

4.2 Dataset

There are not many datasets publicly available which are truly based on wildlife analysis, out of which the main dataset which is going to be applied in this project is the Lindenthal Camera Traps Dataset. This dataset is publicly available at zero cost provided by LILA BC. This dataset is an ideal choice for option 2 according to the project brief, as the data inside this contains all essential things like 3D feature extraction, re-identification, and real depth data. This dataset comprises 775 video sequences captured at Lindenthal Wildlife Park in Cologne, Germany. Intel RealSense D435 stereo camera has been used to capture the data. This dataset supplies all mandatory requirements like synchronised colour, infrared, and real-time depth image streams.

There is a subdivision of this dataset available which contains 412 annotated images, including 1,038 individual animal annotations. Further, this consists of instance masks, bounding boxes, class labels, and track IDs enabling individual re-identification across video sequences. Animals like goat, deer, and many birds have been captured and saved in this database in both day and night time intervals.

This dataset is best considered as an alternative for option 2 of the dissertation project brief, as it is publicly available and facilitates both 3D feature extraction and re-identification. Another database which may be required as a backup option of LILA is SA-FARI dataset. This will only be used if the primary dataset fails to provide enough information.

4.3 Pre-Processing

Data pre-processing is the process of converting data into a subsampled format for a short time after it gets documented by Robot Operating System. Basically, the main goal of performing this process is the reduction of redundancy and computational consumption. On the other hand, MegaDetector v5 will be applied to refine data from empty spaces and to produce bounding boxes that exactly match animals.

Well-tuned positioning of animals will be generated using a Mask R-CNN model, which will produce segmentation masks for each animal detection.

Depth values which are close to object boundaries are mostly loud in noise; hence, to resolve this problem, a 5×5 median filter is applied. This will remove outliers and improve depth reliability. Along with this, data expansion methods like horizontal flipping and brightness adjustment are used on the training set to improve model generalisation. This will also prevent data leakage and ensure fair evaluation.

4.4 3D Feature Extraction

There are two methods which must be applicable to get fine numerical 3D features. The first is a pose estimation model which pinpoints main body spots of animals such as the spine and joints. Then it comprises into 3D coordinates to form a skeleton depiction. Another method is used to transform animal pictures into 3D coordinates and process them using PointNet++ to capture whole shape features.

4.5 Model Development and Evaluation

A Siamese network using triplet loss will learn to identify animals by combining 2D image features from ResNet50 with 3D shape features from PointNet++. For welfare analysis, an LSTM model will analyse sequences of 3D keypoints to detect unusual posture or movement. The models will be built in Python using PyTorch and

trained on Google Colab Pro. The results will be compared with 2D-only models to show how much improvement comes from using 3D features.

5. Analysis of Risks and Ethical Issues

5.1 Technical Risks

The primary risk of this research is quality of depth data obtained from Intel RealSense D435 not in controlled location but in outdoor territory, where elements like long distances, and changing lighting can create huge impact on outcomes. This issue can be overcome by two solutions: applying median filtering and ignoring findings which are inappropriate.

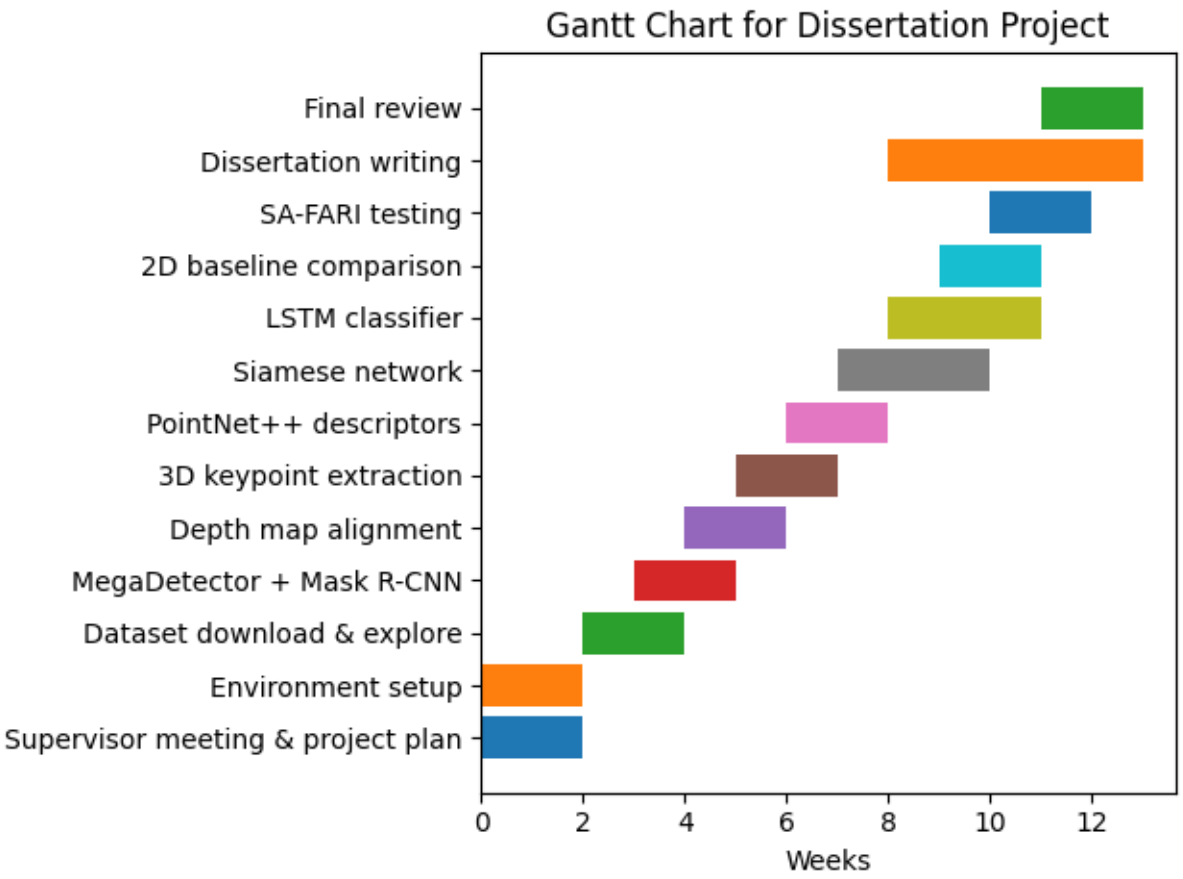
Another problem which may arise is limited size of the Lindenthal dataset; this may cause overfitting while training models.

5.2 Ethical Risks

This study only uses datasets which are publicly available; there is no any direct interaction with live wild animals. And this project is fully automated; hence factors such as fieldwork and camera setup would not be a requirement. Hence this study does not require any permission from the University of East Anglia Ethics Committee.

6. Project Work Plan and Gantt Chart

Phase	Task	Start	End	Duration
1-Setup	Supervisor meeting	Week 1 19-May	Week 1	1 week
1-Setup	Environment setup	Week 1 19-May	Week 2	2 weeks
2-Data	Lindenthal dataset download & exploration	Week 2 26-May	Week 3	2 weeks
2-Data	MegaDetector v5 blank frame filtering	Week 3 2-Jun	Week 4	2 weeks
2-Data	Mask R-CNN segmentation pipeline	Week 3 2-Jun	Week 5	2 weeks
2-Data	Depth map alignment & median filtering	Week 4 9-Jun	Week 5	2 weeks
3-Features	3D keypoint extraction	Week 5 16-Jun	Week 6	2 weeks
3-Features	PointNet++ shape descriptor extraction	Week 6 23-Jun	Week 7	2 weeks
4-Models	Siamese network re-ID model	Week 7 30-Jun	Week 9	3 weeks
4-Models	LSTM welfare classifier	Week 8 7-Jul	Week 10	2 weeks
5-Evaluation	2D baseline comparison	Week 9 14-Jul	Week 10	2 weeks
5-Evaluation	SA-FARI generalisation testing	Week 10 21-Jul	Week 11	2 weeks
6-Writing	Dissertation writing	Week 8 7-Jul	Week 13	6 weeks
6-Writing	Final review	Week 12 4-Aug	Week 13	2 weeks
7-Deadline	Dissertation submission	20-Aug	—	—



7. Structure and Technical Writing Style

This dissertation consists of 7 different sections, following a standard academic structure for postgraduate study research. The very first section is introduction which covers limitations of 2D camera trap imaging for wildlife monitoring and explains the motivation for combining stereo depth sensing with deep learning. After that, the second section is aims and objectives which contain clear research goals aligned with the project brief. Then the literature review section outlines key areas, including limitations of 2D systems, advances in animal re-identification, depth-based welfare analysis, and the move toward 3D feature integration. Methods and methodology provides brief information about a four-stage pipeline: data preprocessing, 3D feature extraction, model development, and evaluation. Risks and ethical considerations have also been discussed.

8. References

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